Hemanth Kandimalla

Applied AI Scientist | Agent Alignment & Control **Portfolio** hemanthiitj.github.io/Hemanth/

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Research focus: Alignment and control of agentic models through harness design, continuous memory and context, and graph-based execution.
AI engineering: 3+ years building domain-adapted models, training/serving infrastructure, and verifiable tool workflows. Author of technical books and the Neural Atlas journal.

PROFESSIONAL EXPERIENCE

Celebal Technologies Sep 2026 – Present
Senior Software Engineer – Data Science | Applied AI Scientist Bengaluru, India

- Investigate **agent alignment and control** across continuous memory and context, workflow/dependency graphs, and memory/knowledge graphs.
- Engineer domain-specialized SLM/LLM applications and agent workflows spanning model adaptation, evaluation, inference efficiency, and scalable deployment.

C-DAC (Centre for Development of Advanced Computing) Aug 2023 – Aug 2026
Knowledge Associate | Applied AI Research Bengaluru, India

- Delivered **10 AI proofs of concept**. Adapted a **3B-parameter LLM** with retrieval/reranking, grounded generation, memory, safety controls, and FastAPI/vLLM serving.
- Engineered **agent harnesses** with persistent memory, context assembly, MCP/JSON-RPC tools, and bounded **plan–execute–verify–repair loops** with permission gates and rollback.
- Profiled training and inference through TTFT, TPOT, throughput, and GPU-memory analysis to guide KV-cache, batching, quantization, and precision decisions.
- Built an adversarial-robustness module with red-team/blue-team agent loops, prompt-injection and jailbreak tests, groundedness checks, and regression gates.

MediaTek | Modem Software Engineer Intern Jul 2022 – Jan 2023

SELECTED OPEN SOURCE

TrainScale YAML-driven PyTorch training framework. Built DPO/SimPO objectives, LoRA/QLoRA workflows, DDP/FSDP execution, and restartable checkpoints. Integrated Triton autograd kernels for cross-entropy, RMSNorm, and RoPE with torch.compile/CUDA Graph execution paths.

SDKAtlas AST-based Python SDK introspection without importing target packages. Resolves signatures, inheritance, and resource graphs into 14 CSV/JSONL/SQLite artifacts.

RESEARCH & TECHNICAL WRITING

Agentic State Technical monograph `https://neuralatlasai.github.io/AutonomyAtlas/`

The Model Stack Book/atlas, in development `https://neuralatlasai.github.io/The_Model_Stack/`

Engineering Reliable Agentic AI Systems Book project
`https://neuralatlasai.github.io/Engineering-Reliable-AgenticAI-Systems/`

Neural Atlas AI Engineering Journal
`https://neuralatlasai.github.io/neural-atlas-ai-engineering-journal/`

TECHNICAL SKILLS

Harness & loop engineering State machines, budgets, stop rules, MCP, JSON-RPC, Agent Skills
Context engineering Continuous memory, continuous context, persistent state, RAG
Graph engineering Workflow/dependency graphs, memory/knowledge graphs
Models & training Python, PyTorch, Transformers, SFT, DPO/SimPO, LoRA/QLoRA, DDP/FSDP
GPU & serving Triton, CUDA Graphs, torch.compile, vLLM, KV-cache, quantization, FastAPI

EDUCATION

Indian Institute of Technology Jammu 2021 – 2023
M.Tech, Communications and Signal Processing CGPA: 8.34/10